

Effect of cure shrinkage of epoxy molding compound on warpage behavior of semiconductor package

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ABSTRACT

Cure shrinkage of the epoxy molding compound (EMC) for semiconductor package during molding process, was precisely measured using a fiber Bragg grating (FBG) sensor by measuring a Bragg wavelength (BW) shift. A dielectric sensor was simultaneously used to monitor the curing process of the EMC and define its gelation point. The relationship between internal strain development of the EMC and its degree of cure was investigated by correlating two measured values (The internal strain from the FBG sensor and the degree of cure from the dielectrometry). The viscoelastic properties of the EMC were also measured by a three-point bending stress relaxation test. Finally, the warpage of EMC-aluminum bi-layer strip was measured with digital image correlation (DIC) method and compared with the result from the finite element analysis. As a result, it was confirmed that the cure shrinkage of the EMC should be considered as an important factor in predicting warpage behavior of the semiconductor.

1. Introduction

The demand for semiconductors is increasing with the advent of the fourth industrial revolution such as AI, self-driving car, IoT, and 5G industry [1,2]. The semiconductor market has required fast data processing speed and high-density storage [3]. In addition, the rapid growth of wearable devices has required thinner and smaller semiconductor packages [4]. However, semiconductor packaging technologies have faced several technical limitations [5]. Among the various issues related to packaging technology, the warpage of the package is one of the major concerns. It causes a variety of defects in the surface mount technology (SMT) process such as inducing mounting defect when mounting a chip or a poor contact of solder joints between the package and printed circuit board (PCB) [6,7]. Since these problems can lead to a critical problem of product reliability degradation, it is essential to predict warpage accurately at the product design stage of electronic devices [8].

Finite element method (FEM) has been used to predict the warpage for improving the product quality [9]. The semiconductor device has

various materials such as metal, polymer, silicon wafer, organic compound, and inorganic compound [10]. The structural warpage with these different materials is mainly caused by the mismatch of the material properties such as modulus and coefficient of thermal expansion (CTE). Therefore, it is essential to determine and reflect the exact material properties of each component to accurately predict the warpage using the FEM.

In recent studies to predict the package level warpage, there is a common feature that warpage is mainly influenced by the thermal properties of materials [11]. In particular, thermal properties of the epoxy molding compound (EMC) such as CTE, viscoelasticity, and the cure shrinkage (CS) are dominant factors in the warpage behavior [12]. Among them, the CS is so complicated because thermoset polymers such as the EMC has chemical and thermo-mechanical coupled behavior with crosslinking of molecules during curing process [13]. Although there is a mold shrinkage provided by the manufacturer, it is an ineffective value for predicting accurate warpage because it is overestimated than the extent to which it affects the actual warpage [14,15]. Furthermore, as

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semiconductors are increasingly miniaturized, the effect of the CS becomes greater. Therefore, the CS behavior should be precisely measured and considered in the packaging process for more accurate warpage prediction.

The CTE and viscoelasticity can be measured using thermomechanical analyzer (TMA) and dynamic mechanical analysis (DMA). However, there is no standard method to measure the CS because of its complicated behavior. Some research for investigating the CS have been conducted using volumetric measurement such as plunger type dilatometer or fiber Bragg grating (FBG) sensor [16,17]. Since the gelation point cannot be sufficiently investigated by volumetric measurement alone, the curing process was not thoroughly analyzed. The gelation point is the moment when the system loses fluidity and becomes very viscous. Therefore, the modulus of the polymer is assumed as zero before reaching the gelation point [18,19], and the shrinkage originating from crosslinking before the gelation point does not induce residual stress in the package, which does not contribute to the generation of the warpage [19,20]. Consequently, a detection of the gelation point is essential to obtain effective CS, which is the shrinkage accumulated after the gelation point. In addition, the gelation point is fundamentally related to the change of molecular structure. Thus, an additional analysis is required to detect the gelation point instead of the volumetric measurement.

In this study, the CS of the EMC was measured using the FBG sensor under similar conditions with the actual packaging process. Also, the dielectric sensor was used to monitor the changes in molecular structure of the EMC during molding process. To verify experiment results by dielectric sensor, the strain development and the degree of cure of the EMC obtained from the FBG sensor and differential scanning calorimetry (DSC) were analyzed. Additionally, the viscoelastic property of the fully-cured EMC was obtained by conducting a three-point bending stress relaxation test. Finally, the FEM model of bi-material strip layer considering the CS and the viscoelastic property of the EMC was developed for predicting the warpage behavior during the molding and the post-curing process. The result from FEM simulation was verified with the warpage measurement result by the digital image correlation (DIC) method.

2. Experiment and background

2.1. Evaluation for effective cure shrinkage of EMC

In this research, the dielectric sensor was used to determine the gelation point of the EMC during curing process. The detected gelation point was used to measure the effective CS with the FBG sensor. The fundamental principle was explained with process of the measurement as following. As shown in Fig. 1 (a), the FBG sensor consists of an optical fiber and a cladding surrounding it. In optical fiber core, a periodic

Bragg grating with refractive index is engraved, which can reflect a particular wavelength of light and transmit all others. The specific wavelength of light reflected by the FBG sensor can be expressed in Bragg wavelength (BW, λ_B). The BW is determined by the refractive index and the distance between the Bragg gratings, which are affected by the condition of the temperature and deformation as shown in Fig. 1 (b). The BW shift from λ_B to λ'_B with temperature and stress-induced deformation can be expressed as,

$$\lambda_B - \lambda'_B = \Delta\lambda_B = \Delta\lambda_B^i + \Delta\lambda_B^d \quad \text{Eq. (1)}$$

where $\Delta\lambda_B$ is amount of BW shift, $\Delta\lambda_B^i$ is called *intrinsic* BW shift related to the temperature change, and $\Delta\lambda_B^d$ is called *deformation* BW shift related to stress-induced deformation [21].

To investigate the strain of the EMC during cure process, it is required that the influence of the temperature ($\Delta\lambda_B^i$) is removed from the measured BW shift ($\Delta\lambda_B$). Thus, BW shift ($\Delta\lambda_B$) according to ambient temperature was measured in advance with a bare FBG sensor to calibrate *intrinsic* BW shift ($\Delta\lambda_B^i$) as shown in Fig. 2. Since $\Delta\lambda_B^i$ has a quadratic relationship with temperature change [22,23], it was obtained as a quadratic function of temperature as following,

$$\Delta\lambda_B^i = 0.00856 \cdot T + 1.186 \times 10^{-5} \cdot T^2 \quad \text{Eq. (2)}$$

where T is temperature in °C unit.

In the experiment, the granular type EMC with 85 wt% silica fillers was filled in the mold and cured with cuboid shape as shown in Fig. 3

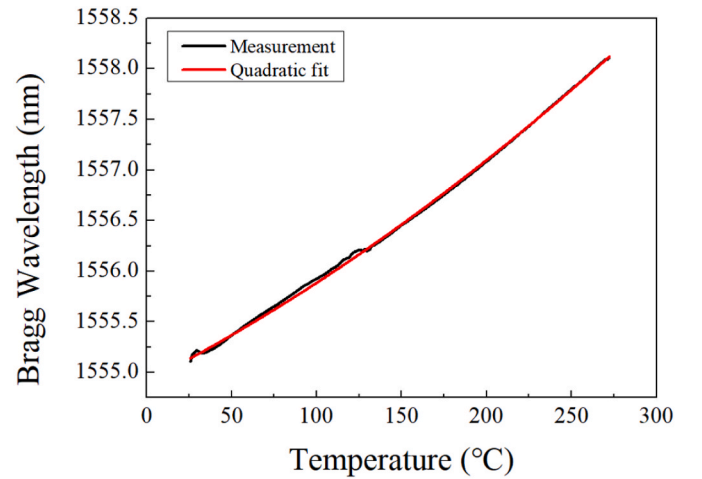


Fig. 2. Bragg wavelength shift according to ambient temperature.

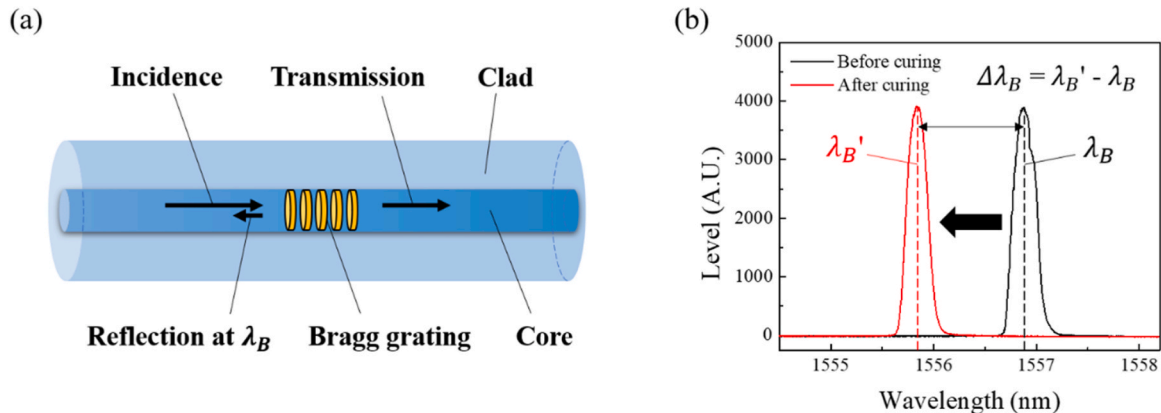


Fig. 1. (a) Schematic of an FBG sensor and (b) a Bragg wavelength shift.

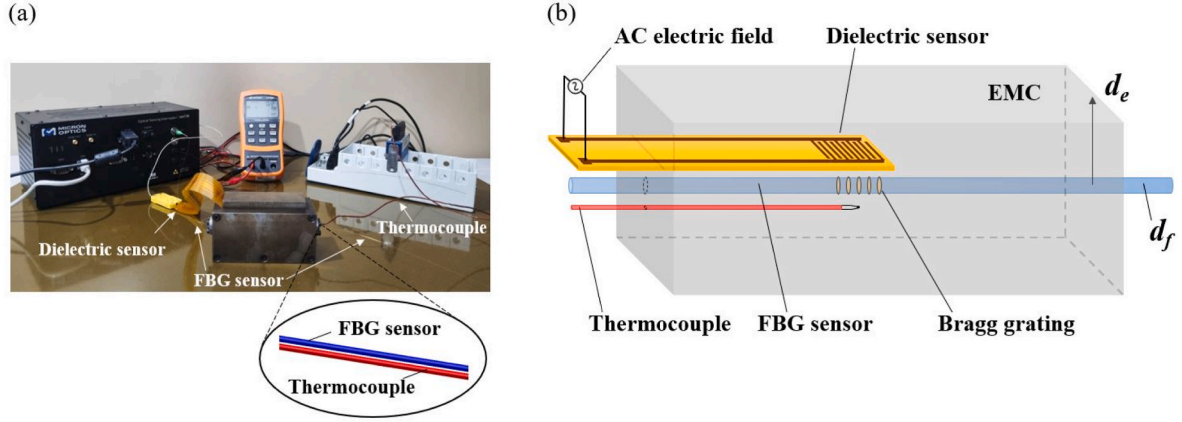


Fig. 3. (a) Experimental setup and (b) schematic of the EMC with the FBG sensor, the dielectric sensor and the thermocouple.

(a). The FBG sensor, the thermocouple and the dielectric sensor were embedded at the center of the EMC to measure the deformation, temperature, and cure status of the EMC during curing process, respectively. After measuring, the $\Delta\lambda_B^d$ related to deformation of the EMC can be obtained by subtracting the $\Delta\lambda_B^i$ from the BW shift.

$$\Delta\lambda_B^d = \Delta\lambda_B - \Delta\lambda_B^i \quad \text{Eq. (3)}$$

As shown in Fig. 3 (b), the radius of the FBG sensor (d_f) was 75 μm and the length of EMC from center to edge (d_e) was 15 mm. In case that the d_e is greater than the d_f as 200 times, it could be assumed that only an uni-axial loading is acting along the fiber longitudinal direction [24]. Then, the strain of the EMC (ϵ) can be determined as [24].

$$\epsilon = \frac{\Delta\lambda_B^d}{\lambda_B(1 - P_k)} \quad \text{Eq. (4)}$$

$$P_k = -\frac{n^2}{2} \{P_{12} - \nu_f(P_{11} + P_{12})\} \quad \text{Eq. (5)}$$

where P_k is the equivalent strain-optic constant, which is consisting of the effective refractive index of the fiber core and cladding materials (n), strain optic constants of Bragg grating (P_{11} and P_{12}), and the Poisson's ratio of the fiber (ν_f) [23]. The P_k is a unique optical property of the FBG sensor, and a reference value of 0.202 was used in this experiment [25].

After the EMC went through the viscous liquid state and has been cured until its molecular weight reaches a maximum, its stiffness starts to rise with increasing crosslink density beyond the gelation point [26]. Therefore, the gelation point should be determined to obtain the

effective CS. In this study, the dielectrometry for monitoring the cure status of polymeric materials was employed to detect the gelation point of the EMC [27]. The dielectric sensor has two interdigitated electrodes on a polyimide substrate, which is embedded in the EMC and connected to an alternating electric field as shown in Fig. 3 (b). Then, the cure status of the EMC was analyzed using an LCR meter (U1732C, Agilent Technologies, USA) with an alternating current of 1000 Hz frequency [28].

As shown in Fig. 4, the temperature profile for curing the EMC was applied similarly to the actual semiconductor package molding process. The EMC curing process consists of two steps; 1) molding process to encapsulates the substrate with gel state EMC in the mold, 2) post-curing process to stabilize the cured EMC by completing the crosslinking process following molding process. In this study, during steps ① and ② of the molding process (see Fig. 4), the compression molding was conducted by applying a pressure of 11 kPa for an hour. Simultaneously, the whole molding jig was bagged with a vacuum bagging film to prevent the pore inside the EMC. After molding, the mold was removed at the initial of step ③. After the cooling step (step ③ of Fig. 4), the post-curing process (step ④) was implemented using the thermal chamber (SH-ND054, SAMHEUNG ENERGY). During the whole process, the strain change in the uniaxial direction of the EMC was monitored through the BW shift from the FBG sensor.

2.2. Evaluation for viscoelastic property of EMC

Since the EMC shows a viscoelastic behavior under the deformation as a polymer, the viscoelastic modulus should be considered in the

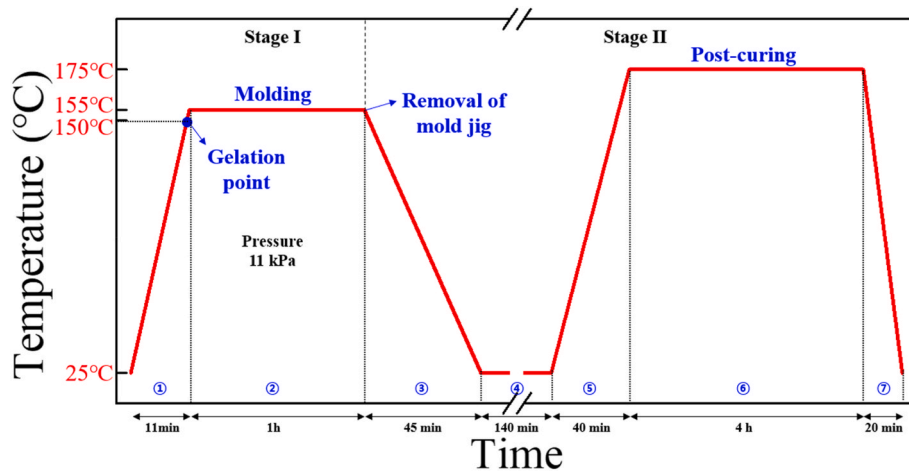


Fig. 4. Temperature profile for measuring the cure shrinkage of the EMC.

warpage simulation. As shown in Fig. 5 (a), the specimen was with dimension of 36 mm × 45 mm × 1.2 mm (span (L) × width (b) × thickness (d)) prepared based on ASTM D-790 under temperature and pressure conditions same with the CS measurement (see Fig. 4). The span-to-depth ratio used in this research was 30, which was the sufficient condition for convergence of flexural modulus due to pure bending in the specimen [29]. The relaxation modulus, $E(t)$, of the EMC was determined through a stress relaxation test using three-point bending with a micro-tensile tester (LLOYD Instrument, USA) and bending test jig based on ASTM E-328 as shown in Fig. 5 (b). The flexural modulus can be obtained using the following equation.

$$E_f = \frac{L^3 m}{4bd^3} \quad \text{Eq. (6)}$$

where E_f is the flexural modulus, and m is the gradient of the initial straight-line portion of the load-deflection curve as shown in Fig. 5 (c). Then, the stress relaxation test was conducted applying deflection of 0.45 mm which is 0.25% flexural strain [30,31]. The deflection was applied to the centerline of the specimen under static temperature conditions. The measurement temperature condition was varied from 25 °C to 205 °C.

2.3. Characterization

The DSC is a common method to obtain the degree of cure of thermosetting polymer which is calculated using the heat generated during crosslinking reaction [32]. In this study, DSC Q20 (TA Instruments, USA) was used to measure the degree of cure of the EMC during the temperature ramp up for molding. The EMC sample was cured under dynamic scanning from room temperature to beyond the molding temperature of 155 °C by applying a ramping rate of 11.8 °C/min same to the ramping rate used when measuring the CS.

2.4. Warpage evaluation of bi-material strip

The EMC-aluminum bi-material specimen for the warpage experiment was prepared with the molding condition of Fig. 4 and dimension as shown in Fig. 6 (a). The molded specimen was subjected to a speckle pattern on top surface of the EMC by spraying to produce a random distribution of gray intensity levels for measurement by the DIC method as shown in Fig. 6 (b). The deformation of the specimen in the high-temperature chamber was monitored during the post-curing process with the DIC method using a modular 3D camera system (ARAMIS Adjustable, GOM) as shown in Fig. 6 (c). The warpage of the bi-material specimen was quantified by vertical deflection of the center point on the top surface of the EMC.

3. Warpage simulation

3.1. Warpage simulation considering cure shrinkage

To predict the warpage behavior of EMC-aluminum bi-material strip, a three-dimensional FEM considering the CS and viscoelastic properties of the EMC was made using ABAQUS 6.14 (Hibbitt, Karlsson and Sorensen, USA). As shown in Fig. 7 (a), the model is composed of an EMC layer and aluminum layer of which thickness are 0.35 mm and 0.2 mm, respectively. Finite-element meshes for the EMC and aluminum were the reduced integration type element (C3D8R). To prevent hourglassing issues of C3D8R mesh type, the 27,000 elements were densely modeled as five elements were set in the thickness direction and a 1 mm of mesh size was set in the plane. The EMC and the aluminum were assumed to be tied. Gravity was applied during the entire simulation.

It was assumed that there is no stress in the bi-material strip before the gelation point because the elastic modulus of the EMC were not developed to affect the residual stress before the gelation point. The initiation point of the warpage simulation was set to the gelation point. In this work, the two-stage simulation was proposed to predict the warpage of the bi-material strip (see Fig. 4). At stage-1, all surfaces were constrained except the top surface of the EMC which was subjected to a pressure of 11 kPa by the mold jig from the gelation point to the end of high-temperature molding curing process (see stage-1 in Fig. 7 (b)). After stage-1, as the mold was removed, the warpage of the bi-material strip was induced from the residual stress due to the change of boundary condition. Then, it was used as the initial warpage state for stage-2. In stage-2, the displacement boundary condition was applied at three corner points of the specimen as shown in Fig. 7 (b). At the point P_1 , the displacement in x -, y -, and z -direction were fixed ($U_1=U_2=U_3=0$). At the points P_2 and P_3 , y - and z -direction ($U_2=U_3=0$), and z -direction ($U_3=0$) displacements were fixed, respectively. The maximum warpage during the curing process was calculated at a center node on the top surface of the EMC in the z -direction displacement.

4. Results and discussion

4.1. EMC properties evaluation

The DMA and rheometer test are commonly used to determine gelation point of polymer materials by analyzing changes in mechanical properties [19]. However, a clear definition of the gelation point using mechanical properties has not yet been established [33]. Since the EMC is a composite material consisted of silica fillers and epoxy binder, it is more difficult to detect the gelation point due to the influence of a high content of silica filler [34,35]. Therefore, in this study, the gelation point was measured and defined using dielectrometry, which is one of the methods to detect the gelation point of the polymer and enables in-situ monitoring [36].

During the curing process, the EMC with an alternating electric field acts as capacitors, and it can be modeled as a parallel circuit of resistance

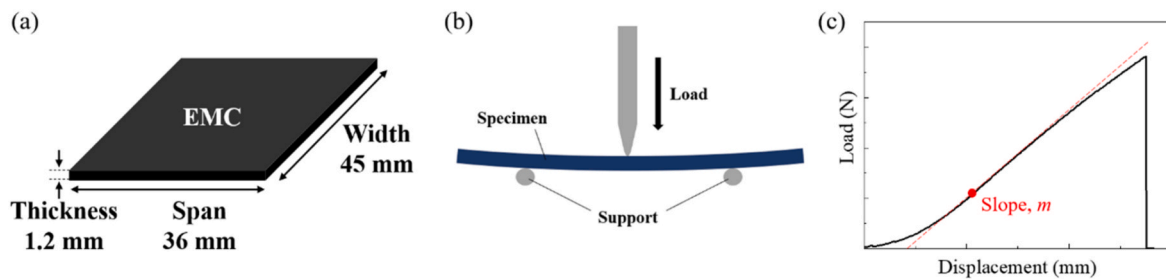


Fig. 5. (a) Dimension of the EMC specimen for three point bending test, (b) schematic of three point bending test, and (c) the slope (m) determination at load-displacement graph.

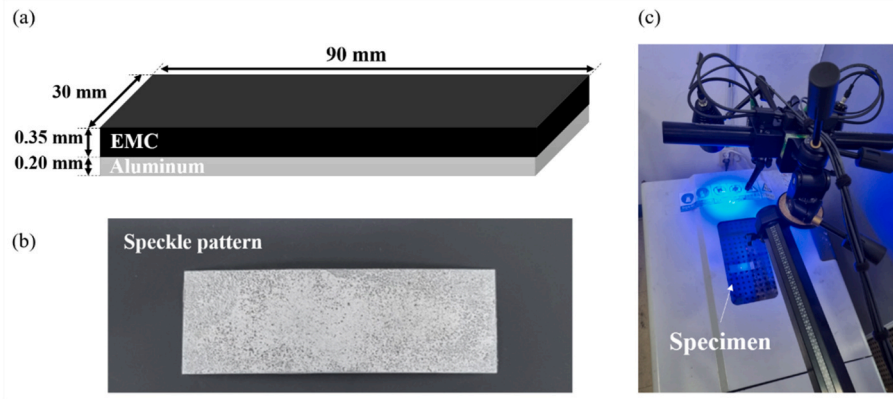


Fig. 6. (a) Schematic of the EMC/aluminum bi-material specimen, (b) the speckle pattern on top surface of the EMC, and (c) DIC experimental setup for warpage analysis.

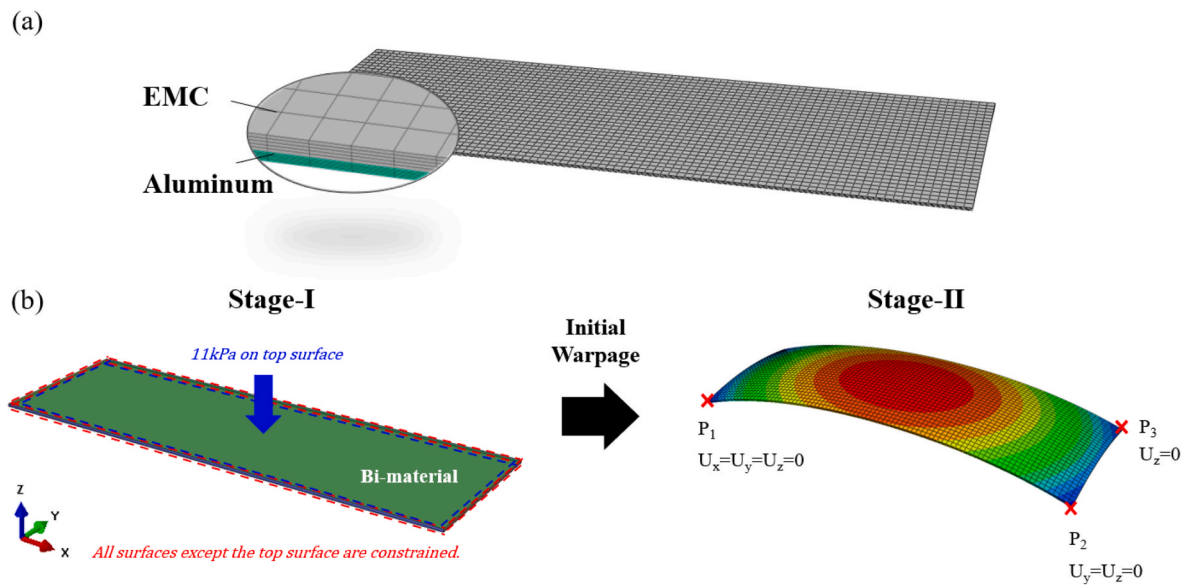


Fig. 7. (a) Geometry consisting of the EMC and aluminum, and (b) schematic of the two-stage simulation for predicting the warpage of the EMC/aluminum bi-material strip using ABAQUS simulation.

(R_m) and capacitance (C_m) as shown in Fig. 8 (a). Since the dipoles and ions in the EMC have high mobility in the viscous liquid state before curing, it shows an ordered array in response to alternating electrical force (see Fig. 8 (b)). On the other hand, their movement is abruptly restricted while EMC becomes gel state or solidifies as the curing reaction progresses, resulting in a relatively disordered array (see Fig. 8 (b)). Therefore, the curing status of the EMC can be grasped through the degree of movement of the dipoles and ions and it can be expressed by

the dissipation factor (D_p), which represents the ratio of the loss energy by the movements of dipoles and ions (E_l) to the energy supplied to the EMC (E_s) [37]. Based on the equivalent circuit model in Fig. 8 (a), the dissipation factor (D_p) can be obtained using Eq. (7) as below [38],

$$D_p = \frac{|E_l|}{|E_s|} = \frac{|I_R \cdot V_m|}{|I_C \cdot V_m|} = \frac{|I_R|}{|I_C|} = \frac{|Z_C|}{|Z_R|} = \frac{1}{\omega \cdot R_m \cdot C_m} \quad \text{Eq. (7)}$$

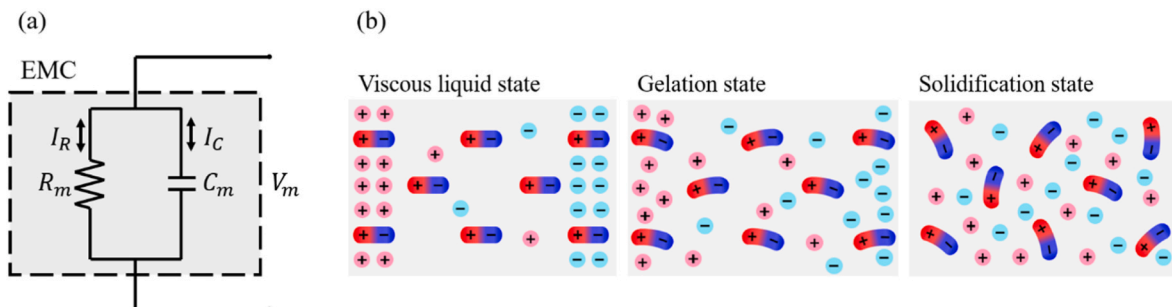


Fig. 8. Schematic of (a) an equivalent electrical circuit and (b) an array of dipoles and ions by electrical force according to the curing state for dielectric material.

where I and Z are the electric current and equivalent impedance, respectively. The V_m is the alternating voltage with angular frequency ω applied to the equivalent circuit, and subscripts R and C represent resistance and capacitance of the equivalent circuit model, respectively.

Fig. 9 (a) shows the dissipation factor (D_p) and the changing rate of dissipation factor (dD_p/dt) during the molding process. The D_p represented the velocity of the dipoles and ions while the dD_p/dt represented the acceleration of their movement. The curing process of the EMC was described as followings. As the EMC in the solid-state was turned into an uncured viscous liquid at the temperature of 60 °C, the dipoles and ions started to move and the D_p also started to increase. While the D_p continued to increase as the dipoles and ions actively moved, the dD_p/dt became a maximum at 112 °C of the curing initiation point [39]. As shown in Fig. 9 (b), the DSC result shows that the EMC began to be cured at 112 °C, which coincides with the curing initiation point (P1) measured by the dielectric sensor. Then, the dD_p/dt started to be decreased as indicated in Fig. 9 (a).

After the initiation of cure, as the polymer chains were crosslinked to form a three-dimensional network, the viscosity of the EMC increased. When the EMC was maximally liquefied, the D_p had the maximum value at which the dD_p/dt crossed zero point (see P2 point in Fig. 9 (a)). At this time, the EMC was in a viscous liquid state, and the dipoles and ions still moved actively to form the arrangement as shown in viscous liquid state of Fig. 8 (b). With the progress of cure, the mobility of the dipoles and ions was gradually restricted by the viscosity of the EMC, which led to the decrease of the D_p .

The point where the dD_p/dt had a minimum value and the D_p was

rapidly decreased, can be defined as the gelation point (P3 in Fig. 9 (a) and (b)) [39]. It was because when all polymer chains were connected to each other, the molecular weight reached to the maximum value (see gelation state of Fig. 8 (b)) [40]. As shown in Fig. 9 (b), it was confirmed that the degree of cure is 0.32 at the gelation point (P3), which is similar to 0.281, known as the general degree of cure of the EMC at the gelation point [41,42].

After gelation point, there was no further molecular weight increase in the EMC, but an increase of crosslink density, thus the viscosity of the EMC was still increased [43]. Therefore, the D_p continued to be decreased further. From the gelation point to the solidification point, the D_p was converged to the minimum value. As a result, it was found that the gelation point detected with the dielectric sensor was 150 °C during the molding process (see P3 point in Fig. 9 (a)).

Using the FBG sensor, the deformation BW shift ($\Delta\lambda_B^d$) of the FBG sensor was measured as shown in Fig. 10 (a). Note that the influence of the temperature on the FBG sensor had been calibrated using Eq. (3) and Fig. 2. Fig. 10 (b) shows the evolution of $\Delta\lambda_B^d$ and dD_p/dt along with the temperature during the molding process. It was observed that the $\Delta\lambda_B^d$ was decreased abruptly near the gelation point (the minimum point of dD_p/dt (150 °C)) (see Fig. 10 (b)). From this result, it was verified again that the dielectrometry is an effective means for detecting the gelation point of the EMC.

By defining the strain state as zero at the gelation point, the strain evolution during the entire process was calculated using Eq. (4), and plotted in Fig. 10 (c). The effective CS at the molding after the gelation

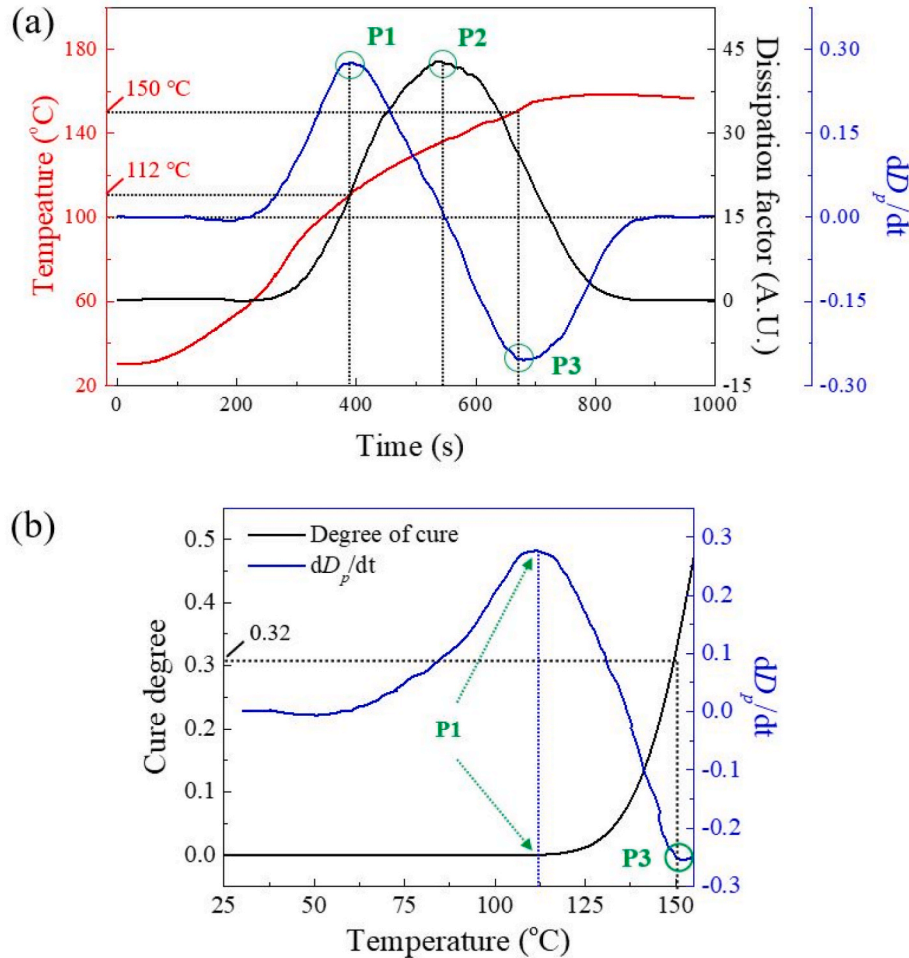


Fig. 9. (a) The measurement result of dissipation factor (D_p) and the rate of dissipation factor (dD_p/dt) during the molding process; The P1, P2, and P3 indicate the curing initiation, the maximum liquid state, and the gelation point, respectively. (b) the DSC result for the degree of cure with the dD_p/dt .

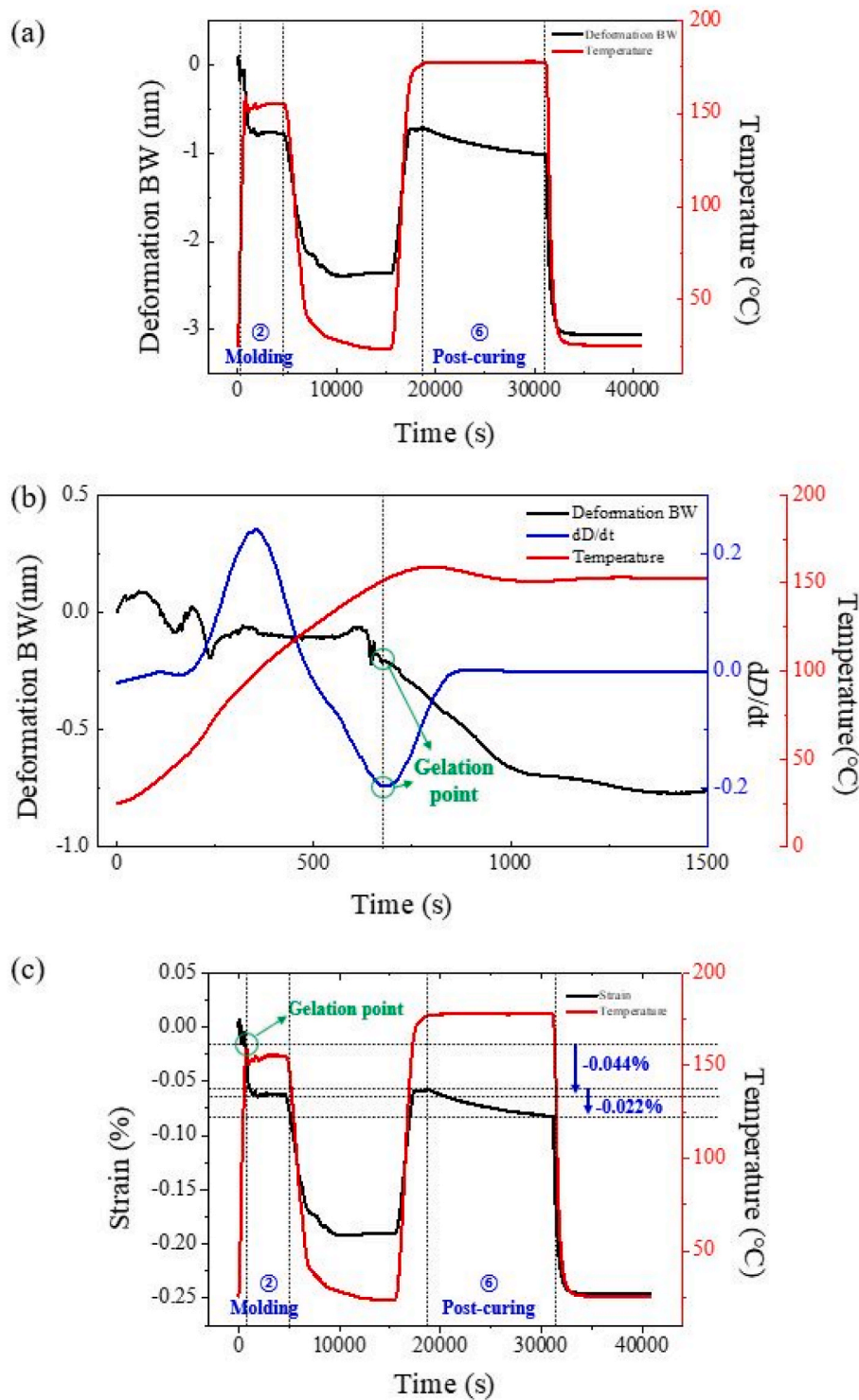


Fig. 10. Experimental results of (a) deformation BW during the molding and post-curing processes, (b) the compared results obtained from the FBG sensor and the dielectric sensor, and (c) strain evolution during whole process with temperature.

point and the post-curing process was obtained, assuming that strain generated before the gelation point as zero. As shown in Figs. 10 (c), 0.044% of the CS of the EMC was generated during the curing process. It is noteworthy that the CS was almost instantly occurred at the beginning of molding process and was not increased any more during remaining molding time (see molding time ② in Fig. 10 (c)). In the post-curing process, the CS of 0.022% occurred in the EMC which was not sufficiently cured in the previous molding process (see the post-mold process time ⑥ in Fig. 10 (c)). Consequently, the 0.044% of effective CS from the

gelation point to the end of the molding process, and 0.022% of post CS during the post-curing process were generated, resulting in a 0.066% of total CS which is similar to the result of 0.077% reported in other studies [44]. Again, these strain values are not negligible and it should be considered to predict accurate warpage of semiconductor package.

The modulus of a thermosetting polymer such as the EMC has a temperature-dependent and time-dependent behavior. Therefore, it is essential to consider the viscoelastic property of the polymer for predicting warpage behavior in the long high-temperature near T_g

packaging process [45]. Fig. 11 (a) shows the time-dependent flexural modulus of the EMC at various temperature conditions obtained from the stress relaxation test. The reference temperature was set as 145 °C near the glass transition temperature (T_g) where the modulus dramatically decreases [46]. The master curve demonstrating the visco-elastic behavior of the EMC was derived by shifting and overlapping the relaxation load curves of each temperature based on the reference load relaxation curve of 145 °C. The time-temperature shift factor (a_T) is expressed by the ratio of relaxation time between the reference (t_0) and the specific temperature (t_s) as below,

$$a_T = \frac{t_s}{t_0} \quad \text{Eq. (8)}$$

where the t_0 and t_s were indicated in Fig. 11 (b). Then, the log-scale time-temperature shift factor can be expressed in terms of the WLF (Williams-Landel-Ferry) equation as below [47],

$$\log a_T = -\frac{C_1(T - T_0)}{C_2 + (T - T_0)} \quad \text{Eq. (9)}$$

where C_1 and C_2 are material constants, T_0 is the reference temperature.

In addition, as shown in Fig. 11 (c), the time-dependent modulus of the EMC can be represented as the Prony series below [47],

$$E(t) = E_\infty + \sum_{i=1}^m E_i \cdot \exp\left(-\frac{t}{\rho_i}\right) \quad \text{Eq. (10)}$$

where t is the time, E_∞ is the long-term modulus, E_i are Prony coefficients, and ρ_i are relaxation times, respectively. The coefficients of the WLF equation and the Prony series were derived from the master curve using a non-linear fitting method, and the calculated results were summarized in Table 1.

The material properties of the EMC and the aluminum used in the simulation are listed in Table 2 [48–51]. The density and Poisson's ratio of each material were considered to be same in entire process. The elastic modulus of the aluminum was set with linear fitted data from 68.08 GPa at 25 °C to 62.08 GPa at 175 °C [51]. Meanwhile, it was assumed that the elastic modulus of the EMC was generated at gelation point as 0.1 GPa, and increased to 6.102 GPa. Since the molding process was carried out for enough time until the EMC was sufficiently cured, it was assumed that when high-temperature molding (step ②) was finished, it would have as much stiffness as the stiffness of fully-cured EMC at the molding temperature (155 °C) [52]. Therefore, the modulus of the EMC was set to increase linearly with time from 0.1 GPa to 0.6 GPa at step ① of Fig. 4 and from 0.6 GPa to 6.102 GPa during at step ② of Fig. 4. Also, the viscoelastic property of EMC was used for the simulation of stage-2 process in Fig. 4 (after molding). The CTE of the aluminum was fixed as 23.1 ppm/°C. On the other hand, the CTE of the EMC were varied as 10.9 and 31.48 ppm/°C before and after the glass transition temperature at which the properties of the epoxy resin are rapidly changed [53]. The CS of the EMC measured during the molding process and the post-curing process from the experiment was assumed to be generated linearly with time during step ② and ⑥ of Fig. 4.

4.2. Warpage analysis

Fig. 12 shows the results of the warpage simulation and the experimental measurement result by the DIC method during the molding and the post-curing processes. Fig. 13 shows the warpage images of the bi-material strip at the end of each step. In actual heating process from 25 °C to 155 °C for molding process (step ① in Fig. 12), the different thermal strain would be generated in the EMC and the aluminum due to the difference in CTE. However, it would not generate any warpage as the bonding interface between the EMC and the aluminum was not accomplished until the gelation point owing to the uncured state of the EMC (they might be freely expanded separately). Therefore, the

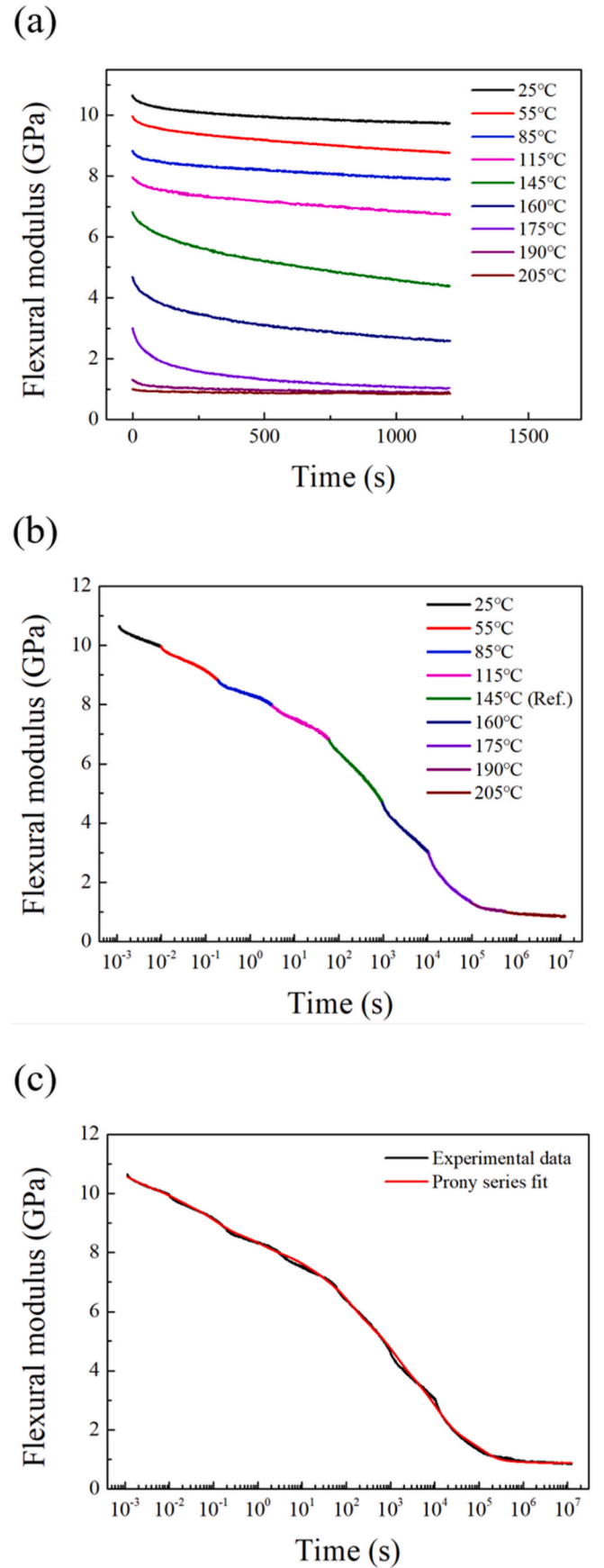


Fig. 11. (a) Stress relaxation test results, (b) master curve of the EMC at a reference temperature of $T_0 = 145$ °C and (c) Prony series fitted curve.

Table 1
Viscoelastic properties of the EMC.

Prony series coefficients		
i	ρ_i	E_i (GPa)
∞	–	0.87602
1	1.0 E-3	0.68028
2	1.0 E-2	0.61268
3	1.0 E-1	0.98347
4	1.0 E+0	0.65519
5	1.0 E+1	0.59788
6	1.0 E+2	1.50489
7	1.0 E+3	1.72125
8	1.0 E+4	2.20321
9	1.0 E+5	1.16786
10	1.0 E+6	0.09549
WLF coefficients		
T_0 (K)	C_1	C_2
418	18.66	500

warpage started to be simulated from the gelation point as shown in Fig. 12. During the heating step in the molding process (step ① in Fig. 4), the thermal strain mismatch between the EMC and the aluminum were generated since the CTE of the EMC is higher than that of the aluminum at this temperature (150–155 °C). This led to the thermal stress that tends to make the bi-material strip into a crying shape.

During the curing process at 155 °C temperature in the molding process (step ②), the EMC was cured and contracted due to the CS while the aluminum was not shrank owing to constant temperature. This affected the strain of the bi-material strip in the smile shape warpage. However, the bi-material strip was enclosed in the mold jig, so the warpage was not generated yet. Eventually, when the mold restraint was released, the warpage of -1.406 mm (smile shape) was predicted from the FEM analysis as shown in the end of step ② of Fig. 13. Note that it was not possible to measure experimentally the warpage because it was

not possible to detach the specimen from the mold followed by speckling preparation for 3D DIC measurement instantly in the end of step ②. Therefore, only FEM simulation result was shown in Fig. 13 in the end of step ②.

In cooling process (step ③ in Fig. 12), the direction of the warpage was changed to the crying shape owing to T_g . The CTE of the EMC is higher than that of the aluminum above the T_g , while the CTE of the EMC is lower than that of the aluminum under the T_g . Therefore, the warpage was generated in negative direction (to be crying shape) from the removal of the mold restraint to reaching to the T_g (146 °C). Contrarily, from the T_g to 25 °C, the warpage was accumulated reversely to the positive direction (to be smile shape). This phenomenon near T_g was indicated in the FEM simulation as the warpage overturn point in Fig. 12 (see the warpage overturn points with green circle in Fig. 12). Finally, the warpage of the bi-material strip became 2.795 mm (crying shape) at 25 °C in the end of step ③ similarly to the FEM simulation result (2.748 mm) as shown in Figs. 12 and 13.

Although the temperature maintaining step at 25 °C (step ④) was progressed for long time, the warpage obtained from the simulation was changed from 2.748 mm to 2.744 mm, which was negligible because the viscoelastic effect of the EMC is too small at room temperature [45]. Similarly to the simulation result, only few warpage changes of 0.003 mm (from 2.795 mm to 2.792 mm) were observed from the experimental measurement using the DIC method (see the end of step ④ in Figs. 12 and 13).

As shown in Fig. 12, in the heating process (step ⑤) of the post-curing, the warpage of the bi-material strip undergoes again the warpage overturn phenomenon owing to T_g effect on CTE values similarly to the case in cooling process (step ③) as described above. As shown in the simulation result of Figs. 12 and 13, the warpage of the bi-material strip was flipped from 2.744 mm (crying shape) to -0.935 mm (smile shape) due to the difference in the thermal expansion of the EMC and the aluminum. However, in the experimental measurement, a sharp warpage overturn was not clearly observed but the abrupt measurement

Table 2
The material properties of the EMC and aluminum for FEM analysis.

	Density ρ (g/cm ³)	Poisson's ratio ν	Elastic modulus		Coefficient of thermal expansion		Glass transition temperature T_g (°C)
			E_{S1} (GPa)	E_{S2} (GPa)	α_1 (ppm/°C)	α_2 (ppm/°C)	
Aluminum	2.67	0.336	Linear fitting	68.08 @25 °C 62.08 @175 °C	23.1		–
EMC	1.82	0.3	0.1–6.102	Viscoelastic property	10.9	31.48	146

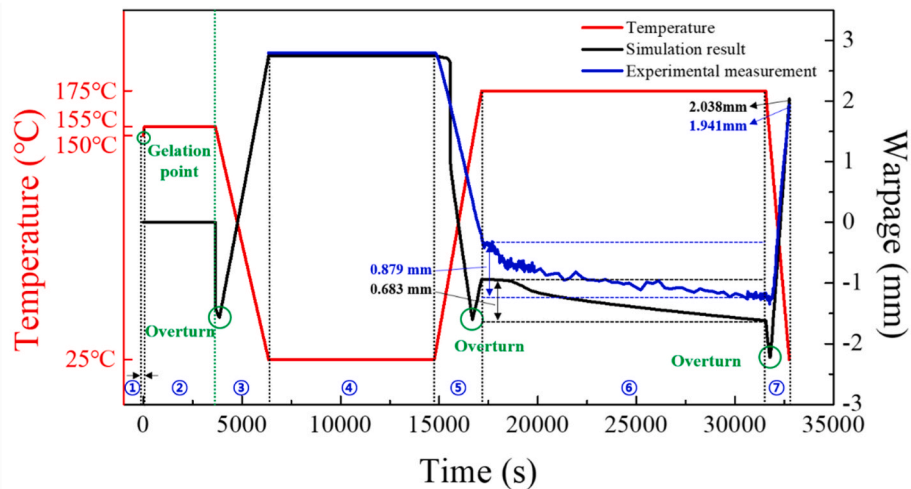


Fig. 12. The result of warpage simulation considering the cure shrinkage and viscoelastic properties of the EMC and warpage experimental measurement by the DIC method during the whole process.

Deformation scale factor : $\times 2.5$						
Step	The end of step ②	The end of step ③	The end of step ④	The end of step ⑤	The end of step ⑥	The end of step ⑦
Temperature profile						
Simulation result	- 1.406mm	2.748mm	2.744mm	- 0.935mm	- 1.617mm	2.038mm
Experimental measurement	-	2.795mm	2.792mm	- 0.188mm	- 1.205mm	1.941mm

Fig. 13. The comparison of the simulation result considering the cure shrinkage and viscoelastic properties of the EMC and experimental measurement result by the DIC method at the end of each step (②~⑦).

oscillation was only observed because the CTE of the EMC gradually changes near the glass transition temperature.

During the temperature maintaining period of the post-curing process (step ⑥), the warpage of the bi-material strip was changed from -0.935 mm to -1.617 mm as a result of the stress relaxation and the CS of the EMC as shown in Figs. 12 and 13. It was also confirmed that the warpage change from the simulation and the experiment were -0.683 mm and -0.879 mm, respectively.

After cooling process of the post-curing (step ⑦), the warpage of the bi-material strip by the simulation was overturned and flipped from -1.617 mm (smile shape) to 2.038 mm (crying shape) due to difference in thermal shrinkage as shown in Figs. 12 and 13. The warpage of the bi-material strip was measured using the DIC method as 1.941 mm with error of 5%. It was also confirmed that all the warpage results predicted by the simulation had a good agreement with the experimental measurement.

To confirm the effect of the CS of the EMC, the simulations without consideration of the CS of the EMC, were conducted and compared with the above simulation result as shown in Figs. 14 and 15 (all other properties were equally applied). After the removal of the mold restraint, the predicted warpage was -1.406 mm with the consideration of

the CS. However, the warpage was predicted as 0.069 mm when the CS was not considered. This warpage difference of 1.475 mm was almost maintained to 1.453 mm until the end of heating process for the post-curing (the end of step ⑤) as shown in Fig. 15.

As shown in Fig. 14, during the post-curing process (step ⑥), it was observed that the warpage of the bi-material strip was changed as -0.683 mm with considering the CS of the EMC, while the warpage was changed as -0.248 mm without considering it. After the final cooling process from the post-curing process (step ⑦), the final warpage was predicted to be 2.038 mm with the consideration of the CS of the EMC and 3.911 mm without the consideration of the CS of the EMC, respectively. As a result, the predicted final warpage results showed about twice difference depending on the CS of the EMC.

From this result, it can be concluded that the CS of the EMC during the molding process (-0.044%) shows an significant effect on the warpage prediction. Fig. 16 (a) shows the warpage change during the post-curing depending on the consideration of the CS of the EMC. It was noteworthy that the effect of the CS of the EMC during the post-curing also had a great effect on the warpage despite the smaller CS (-0.022%) of the EMC during the post-curing than the CS of the EMC during the molding process (-0.044%). It might be due to higher elastic

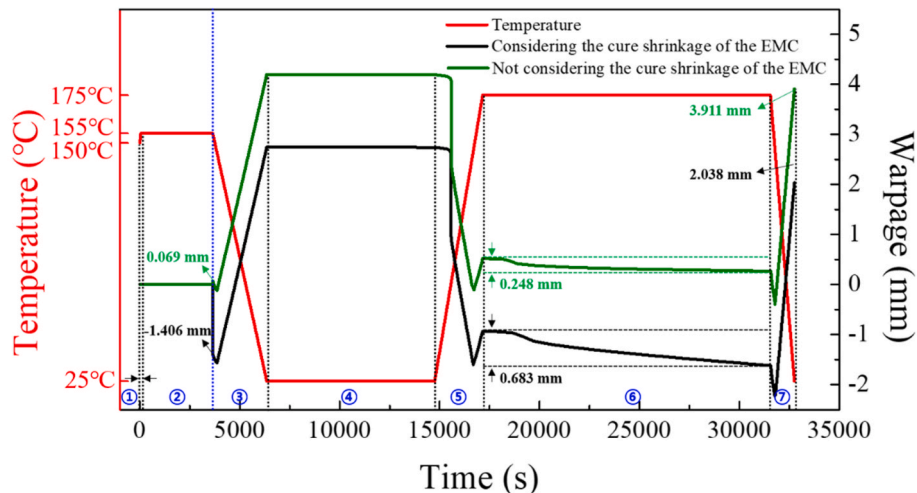


Fig. 14. The comparison of warpage simulation result with and without the consideration in the cure shrinkage of the EMC during whole process.

Deformation scale factor : $\times 2.5$

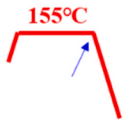
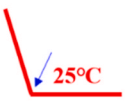
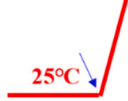
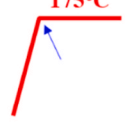
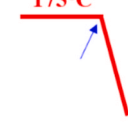
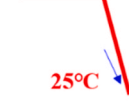
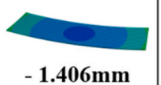
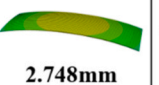
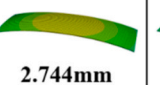
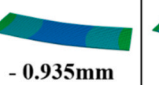
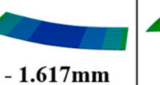
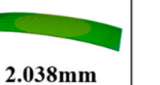
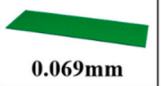
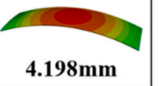
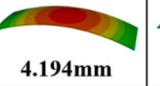
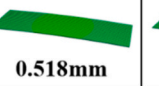
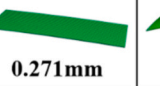
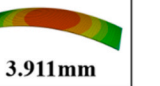
Consideration	Step					
	End of step ②	End of step ③	End of step ④	End of step ⑤	End of step ⑥	End of step ⑦
Cure shrinkage						
O	 - 1.406mm	 2.748mm	 2.744mm	 - 0.935mm	 - 1.617mm	 2.038mm
X	 0.069mm	 4.198mm	 4.194mm	 0.518mm	 0.271mm	 3.911mm

Fig. 15. The comparison of warpage simulation result with and without the consideration in the cure shrinkage of the EMC at the end of each step (②~⑦).

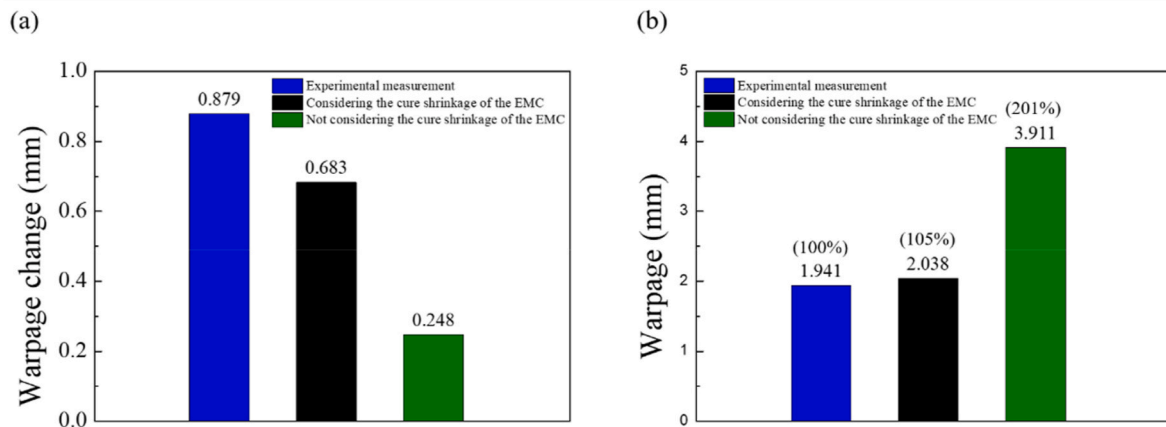


Fig. 16. The comparison result of (a) the change in warpage during the post-curing and (b) final warpage for the experimental measurement by DIC method, and simulations with and without the consideration in the cure shrinkage of the EMC.

modulus in the post curing process than that in the molding process, thus the CS might contribute more to the warpage generation.

As shown in Fig. 16 (b), the total warpage prediction result from the FEM simulation considering the CS of the EMC showed an error with 5% of the experimental measurement. On the other hand, the result from the FEM simulation without the consideration of the CS of the EMC, showed two times higher warpage prediction results than the experimental result. Again, it can be concluded that the effect of the CS is a significant and it should be measured and considered for precise prediction of the warpage of semiconductor package.

5. Conclusion

In this study, the effective CS of the EMC could be precisely measured by using the FBG sensor and the dielectric sensor. The FEM simulation was conducted to predict the warpage of EMC-aluminum bi-material strip considering the effective CS and the viscoelastic property of the EMC and it was compared with the measurement result of the warpage using the DIC method. From the study, it was found that the warpage could be overestimated without consideration of the CS of the EMC. It

can be concluded that the CS of the EMC is an essential factor for accurate warpage prediction in semiconductor packaging.

CRediT authorship contribution statement

Jeong-Hyeon Baek: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Dong-Woon Park:** Writing – review & editing. **Gyung-Hwan Oh:** Methodology, Investigation. **Dong-Ok Kawk:** Resources. **Simon S. Park:** Supervision. **Hak-Sung Kim:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mssp.2022.106758>.

References

- [1] Y.-Q. Chen, B. Zhou, M. Zhang, C.-M. Chen, Using IoT technology for computer-integrated manufacturing systems in the semiconductor industry, *Appl. Soft Comput.* 89 (2020) 106065.
- [2] J. Ahn, T. Ahn, Remeasurement dispatching rule for semiconductor EDS process, in: *International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing*, Springer, 2021, pp. 155–168.
- [3] M. Hochberg, M. Sack, D. Herzog, A. Weisenburger, W. An, R. Fetzner, G. Mueller, A fast modular semiconductor-based Marx generator for driving dynamic loads, *IEEE Trans. Plasma Sci.* 47 (1) (2018) 627–634.
- [4] J. Talledo, Study of the impact of curing condition on flexural strength of a very thin semiconductor package, *Journal of Engineering Research and Reports* (2021) 46–51.
- [5] N.R.E. Lim, E. Arriola, R.L. Moran, J.P. Mercado, R. Dimagiba, J. Gonzaga, A. Ubando, Material properties sensitivity analysis based on thermomechanical analysis on warpage for ball grid array semiconductor packaging, in: *2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, IEEE, 2019, pp. 1–5.
- [6] M. Mohan, M. Ansari, R.A. Shanks, Review on the effects of process parameters on strength, shrinkage, and warpage of injection molding plastic component, *Polym. Plast. Technol. Eng.* 56 (1) (2017) 1–12.
- [7] Y. Sun, H.-S. Lee, B. Han, Measurement of elastic properties of epoxy molding compound by single cylindrical configuration with embedded fiber Bragg grating sensor, *Exp. Mech.* 57 (2) (2017) 313–324.
- [8] D.-S. Choi, Y.-T. Im, Prediction of shrinkage and warpage in consideration of residual stress in integrated simulation of injection molding, *Compos. Struct.* 47 (1–4) (1999) 655–665.
- [9] S.-S. Deng, S.-J. Hwang, H.-H. Lee, Warpage prediction and experiments of fan-out waferlevel package during encapsulation process, *IEEE Trans. Compon. Packag. Manuf. Technol.* 3 (3) (2013) 452–458.
- [10] E.M. Graycocha Jr, F.R.I. Gomez, R.S. Rodriguez, Warpage Mitigation through Diebond Process Improvement with Enhanced Leadframe Configuration.
- [11] E.R. Fitzharris, N. Watanabe, D.W. Rosen, M.L. Shofner, Effects of material properties on warpage in fused deposition modeling parts, *Int. J. Adv. Manuf. Technol.* 95 (5) (2018) 2059–2070.
- [12] M. Zarrelli, I.K. Partridge, A. D'Amore, Warpage induced in bi-material specimens: coefficient of thermal expansion, chemical shrinkage and viscoelastic modulus evolution during cure, *Compos. Appl. Sci. Manuf.* 37 (4) (2006) 565–570.
- [13] P. Chen, Z. Ji, Y. Liu, C. Wu, N. Ye, H. Takiar, Warpage prediction methodology of extremely thin package, in: *2017 IEEE 67th Electronic Components and Technology Conference (ECTC)*, IEEE, 2017, pp. 2080–2085.
- [14] S.-S. Yeh, P.-Y. Lin, S.-P. Jeng, W.-Y. Lin, M.-C. Yew, K.-C. Lee, S.-T. Leu, K.-C. Liu, A novel integrated warpage prediction model based on characterization of viscoelasticity in time domain and chemical shrinkage for molded underfill, in: *2015 10th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT)*, IEEE, 2015, pp. 79–82.
- [15] C. Kim, S. Phansalkar, H.-S. Lee, B. Han, On curing-induced residual stresses after molding processes: mold shrinkage, chemical shrinkage or both?, in: *2019 IEEE 69th Electronic Components and Technology Conference (ECTC)*, IEEE, 2019, pp. 1382–1386.
- [16] S.-S. Yeh, P.-Y. Lin, K.-C. Lee, J.-H. Wang, W.-Y. Lin, M.-C. Yew, P.-C. Lai, C.-K. Hsu, C.-C. Yang, S.-P. Jeng, An integrated warpage prediction model based on chemical shrinkage and viscoelasticity for molded underfill, in: *2018 IEEE 68th Electronic Components and Technology Conference (ECTC)*, IEEE, 2018, pp. 249–254.
- [17] S.P. Phansalkar, C. Kim, B. Han, P.J. Gromala, Volumetric effective cure shrinkage measurement of dual curable adhesives by fiber Bragg grating sensor, *J. Mater. Sci.* 55 (22) (2020) 9655–9664.
- [18] Y. Wang, B. Han, Characterization of effective chemical shrinkage and modulus evolution during polymerization, in: *2007 12th International Symposium on Advanced Packaging Materials: Processes, Properties, and Interfaces*, IEEE, 2007, pp. 145–146.
- [19] A.R. Adli, K.M. Jansen, Numerical investigation and experimental validation of residual stresses building up in microelectronics packaging, *Microelectron. Reliab.* 62 (2016) 26–38.
- [20] Y. Sun, Y. Wang, C. Jang, B. Han, K. Choi, Generalized hybrid modeling to determine chemical shrinkage and modulus evolutions at arbitrary temperatures, *Exp. Mech.* 53 (9) (2013) 1783–1790.
- [21] Y. Wang, L. Woodworth, B. Han, Simultaneous measurement of effective chemical shrinkage and modulus evolutions during polymerization, *Exp. Mech.* 51 (7) (2011) 1155–1169.
- [22] K.J. Kim, A. Bar-Cohen, B. Han, Thermo-optical modeling of polymer fiber Bragg grating illuminated by light emitting diode, *Int. J. Heat Mass Tran.* 50 (25–26) (2007) 5241–5248.
- [23] Y. Sun, Y. Wang, Y. Kim, B. Han, Dual-configuration fiber Bragg grating sensor technique to measure coefficients of thermal expansion and hygroscopic swelling, *Exp. Mech.* 54 (4) (2014) 593–603.
- [24] Y. Wang, B. Han, D. Kim, A. Bar-Cohen, P. Joseph, Integrated measurement technique for curing process-dependent mechanical properties of polymeric materials using fiber bragg grating, *Exp. Mech.* 48 (1) (2008) 107–117.
- [25] A.Y. Rabbi, M.R. Kaysir, M.J. Islam, Numerical modeling to evaluate the performance of FBG-based strain sensors, in: *2019 International Conference on Computer, Communication, Chemical, Materials and Electronic Engineering (IC4ME2)*, IEEE, 2019, pp. 1–4.
- [26] J.K. Gillham, Formation and properties of thermosetting and high Tg polymeric materials, *Polym. Eng. Sci.* 26 (20) (1986) 1429–1433.
- [27] G.M. Maistros, I.K. Partridge, Dielectric monitoring of cure in a commercial carbon-fibre composite, *Compos. Sci. Technol.* 53 (4) (1995) 355–359.
- [28] H.L. Lee, *The Handbook of Dielectric Analysis and Cure Monitoring*, vol. 156, Lambert Technology LLC, Boston, MA, 2014.
- [29] A. Mehandiratta, S. Bandyopadhyaya, V. Kumar, D. Kumar, Experimental investigation of span length for flexural test of fiber reinforced polymer composite laminates, *J. Mater. Res. Technol.* 7 (1) (2018) 89–95.
- [30] Y. Kim, H. Lee, S. Park, X. Zhang, Stress relaxation test of molding compound for MEMS packaging, in: *13th InterSociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems*, IEEE, 2012, pp. 290–296.
- [31] X. Huang, S. Lee, Stress relaxation in plastic molding compounds, in: *Proceedings of the 4th International Symposium on Electronic Materials and Packaging*, IEEE, 2002, pp. 37–42, 2002.
- [32] A. Atarsia, R. Boukhili, Relationship between isothermal and dynamic cure of thermosets via the isoconversion representation, *Polym. Eng. Sci.* 40 (3) (2000) 607–620.
- [33] W. Brostow, S.H. Goodman, J. Wahrmund, *Handbook of Thermoset Plastics: 8. Epoxies*, Elsevier Inc. Chapters, 2013.
- [34] B.-S. Chiou, S.-S. Raghavan, S.A. Khan, Effect of colloidal fillers on the cross-linking of a UV-curable polymer: gel point rheology and the winter–chambon criterion, *Macromolecules* 34 (13) (2001) 4526–4533.
- [35] M. Linec, B. Mušič, The effects of silica-based fillers on the properties of epoxy molding compounds, *Materials* 12 (11) (2019) 1811.
- [36] C. Carlini, P. Rolla, E. Tombari, Measurement method and apparatus for monitoring the kinetics of polymerization and crosslinking reactions by microwave dielectrometry, *J. Appl. Polym. Sci.* 41 (3–4) (1990) 805–818.
- [37] J.S. Kim, D.G. Lee, Measurement of the degree of cure of carbon fiber epoxy composite materials, *J. Compos. Mater.* 30 (13) (1996) 1436–1457.
- [38] K.G. Bang, J.W. Kwon, J.W. Lee, Measurement of the degree of cure of glass fiber–epoxy composites using dielectrometry, *J. Mater. Process. Technol.* 113 (1–3) (2001) 209–214.
- [39] H.S. Kim, Reduction of fabrication thermal residual stress of the hybrid cure-cured structure using a dielectrometry, *Compos. Sci. Technol.* 67 (1) (2007) 29–44.
- [40] W. Brostow, S.H. Goodman, J. Wahrmund, *Epoxies, Handbook of Thermoset Plastics*, Elsevier, 2014, pp. 191–252.
- [41] S. Han, K. Wang, C. Hieber, C. Cohen, Characterization of the rheological properties of a fast-curing epoxy-molding compound, *J. Rheol.* 41 (2) (1997) 177–195.
- [42] S. Han, F.S. Costa, S.R. Ray, Integrated flow analysis during filling and post-filling stage of thermoset molding, in: *AIP Conference Proceedings*, American Institute of Physics, 2004, pp. 180–185.
- [43] K. Dusek, W.J. MacKnight, *Cross-linking and Structure of Polymer Networks*, 1988.
- [44] C. Kim, S.P. Phansalkar, H.S. Lee, B. Han, Measurement of effective cure shrinkage of epoxy-based molding compound by fiber Bragg grating sensor using two-stage curing process, *J. Appl. Polym. Sci.* 51620.
- [45] S. Ronan, T. Alshuth, S. Jerrams, N. Murphy, Long-term stress relaxation prediction for elastomers using the time–temperature superposition method, *Mater. Des.* 28 (5) (2007) 1513–1523.
- [46] S.-H. Chae, J.-H. Zhao, D.R. Edwards, P.S. Ho, Characterization of Viscoelasticity of Molding Compounds in Time Domain, *International Electronic Packaging Technical Conference and Exhibition*, 2009, pp. 435–441.
- [47] G. Hu, A.A. Tay, Y. Zhang, W. Zhu, S. Chew, Characterization of viscoelastic behaviour of a molding compound with application to delamination analysis in IC packages, in: *2006 8th Electronics Packaging Technology Conference*, IEEE, 2006, pp. 53–59.
- [48] T. Tiedje, K. Meier, K. Bock, Experimental methodology for low temperature vibration investigations on electronic modules, in: *2016 39th International Spring Seminar on Electronics Technology (ISSE)*, IEEE, 2016, pp. 237–240.
- [49] K. Nishido, K. Hamaguchi, M. Takekoshi, N. Suzuki, T. Nonaka, Study of influence of mechanical property of temporary bonding adhesive on warpage of FO-WLP, in: *2017 International Conference on Electronics Packaging (ICEP)*, IEEE, 2017, pp. 82–85.
- [50] W. Xia, M. Xiao, Y. Chen, F. Wu, Z. Liu, H. Fu, Thermal Warpage Analysis of PBGA Mounted on PCB during Reflow Process by FEM and Experimental Measurement, *Soldering & Surface Mount Technology*, 2014.

- [51] P. Gudlur, A. Muliana, M. Radovic, Effective thermo-mechanical properties of aluminum–alumina composites using numerical approach, *Compos. B Eng.* 58 (2014) 534–543.
- [52] R. Hardis, J.L. Jessop, F.E. Peters, M.R. Kessler, Cure kinetics characterization and monitoring of an epoxy resin using DSC, Raman spectroscopy, and DEA, *Compos. Appl. Sci. Manuf.* 49 (2013) 100–108.
- [53] G. Hu, J.-E. Luan, S. Chew, Characterization of Chemical Cure Shrinkage of Epoxy Molding Compound with Application to Warpage Analysis, 2009.